

Leo's Auto Database Project

Report 1

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Course and section: COSC 457 - 001

Instructor: Sung-Chul Hong Submission date: [date]

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Manager ownership and progress

Colors identify the manager; labels show progress. Done means completed in this revision. To do means a manager still needs to review, provide or assemble the work.

Manager	Status	Responsibility
Interview Manager Mae	Done	Introduction, interview record, business process and interview answers in section 4. Completed form and signature attached as Appendix B.
Database Manager Marquise	To do	Review data requirements, Chen ERD and schema against the interview; document corrections and attach the design as Appendix A.
Application Manager Marcel	To do	Review application behavior and query priorities; provide two or three screens, confirm tools and setup, and record required changes.
Report Manager Dashaun	To do	Combine the pieces, fill actual dates and contribution log, finalize contents and page numbers, check the assembled report and submit.

Shared work remains with the named managers and the group. The interview evidence is attached; design review, screens, dates, contribution records and final checks still need completion.

1 Introduction

Our project is a small database application informed by an interview with Leo's Auto Inc., the confirmed business at 14 Willow Ave, Towson, MD 21286. The project focuses on the information needed to connect a customer, a vehicle, a repair visit, the work performed, and the resulting bill. We want to understand that process well enough to explain the data model and demonstrate a complete example using fictional records.

Mae Beeman interviewed Brooke Maenner, whose role was recorded as Daughter / Front Desk Manager, in person at the shop on September 29, 2026, at 4:45 PM. The shop uses digital management software for estimates and paper repair orders for mechanics. Finding older paper records for warranty checks and tracking wrong-part returns are difficult. Brooke selected repair-history searches by customer phone number and an open-jobs dashboard, including waiting for parts and ready for pickup, as the prototype's priorities.

Our proposed scope covers customer and vehicle records, repair orders, labor and parts used on a job, invoices, and payments. The interview prioritizes customer repair history and open repair jobs; unpaid bills were not selected as a priority. Customers pay in full by cash or card at pickup, and closed, paid invoices remain unchanged, with corrections handled by a separate credit or refund. Appointments, payroll, online payments, supplier ordering, and a customer portal remain outside the proposed scope.

The design phase will turn the interview answers into data and process requirements. A Chen entity relationship diagram will show the types of information and how they connect. A matching schema will identify the tables, primary keys and foreign keys. We will check rules such as whether a vehicle has one current customer, whether a repair can involve several mechanics, and whether one visit can produce more than one bill. Any difference between the interview and our starting design will be recorded and addressed.

The implementation phase will connect those tables to a small application. A Python and SQLite sample is available as a starting point, but our final tools will be [\[confirmed course tools\]](#). We will use fictional records to test customer entry, a repair with labor and parts, invoicing, payment, and the selected reports. We will also check invalid entries and confirm that saved information remains available after restarting the application.

Work is divided among interview and requirements, database design, application development, and report assembly. Each member will review and explain their own contribution. The final package will include source code, database setup instructions, a functional guide, a step-by-step tutorial, and a demonstration. Our aim is a manageable application whose design, implemented behavior and documentation agree.

2 Interview and target business

Interview Manager: Mae Beeman

Interview record

Date and time: September 29, 2026, 4:45 PM

Location or mode: In person at the shop

Interviewee and role: Brooke Maenner, Daughter / Front Desk Manager

Students present: Mae Beeman

Signed proof: Completed form attached as Appendix B; dated September 29, 2026.

The supplied form includes an electronic signature. It confirms that the interview occurred; it does not mean that the interviewee approved the report or application.

Business profile

Item	Interview answer
Business name and address	Confirmed: Leo's Auto Inc., 14 Willow Ave, Towson, MD 21286.
History and branches	Family-owned and operated; founded by Leo in 1978. One location.
Size and staff	About 5-6 staff. Family members run the front desk; mechanics handle repairs.
Activities and services	General auto repair and maintenance, A/C, brakes and tire repair. Service only; no car sales.
Customers and service area	Towson and the surrounding area. Mainly personal/family vehicles; few fleet accounts.
Total sales	Revenue was not disclosed.
Current tools	Digital shop management software for estimates and invoicing; printed paper repair orders (ROs) for mechanics.

Questions answers and evidence

Appendix B contains the supplied three-page completed interview form, including all questions, answers and the signature. Section 4 identifies the source question for each answer.

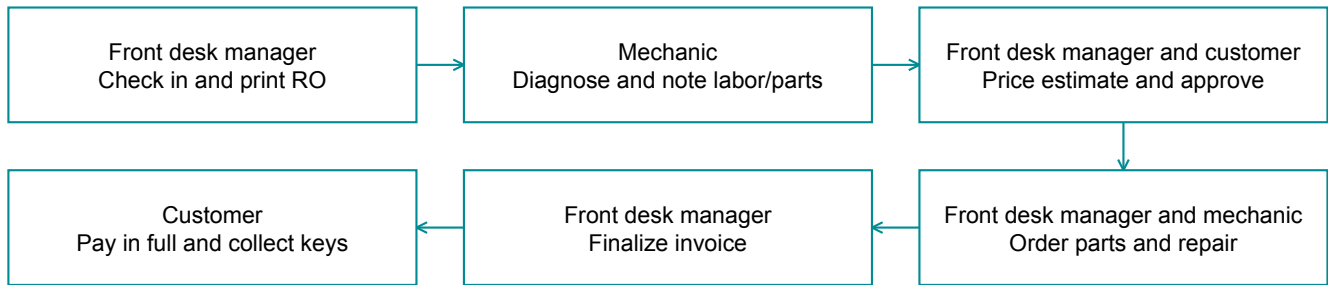
Main findings:

1. Brooke prices estimates and records verbal phone approval before ordering parts.
2. Older paper repair histories and physical returns of wrong parts to NAPA are difficult to track.
3. Phone-number history searches and job status are the two selected priorities.

Permissions: Business name use and note-taking allowed. Exterior/front desk photos discouraged; no plates or personally identifying information.

3 Business process

Confirmed workflow from interview question 3; approval and payment rules from question 7.



One ordinary visit

1. The front desk manager checks in the customer, records contact and vehicle details, the complaint and exact arrival mileage, and prints an RO.
2. A mechanic takes the car and RO to a bay, diagnoses the problem, and writes the required parts and labor on the RO.
3. The front desk manager prices the estimate and calls the customer for approval. Approval is verbal; the name and time are written on the RO.
4. After approval, the front desk manager orders parts and the mechanic completes the repair.
5. The front desk manager finalizes the invoice in the shop system.
6. The customer pays the full amount by cash or card at pickup and receives the keys.

Other activities in scope

Catalog and staff records: The interview did not identify who maintains these records. The front desk manager handles estimates, parts ordering and invoices; mechanics record diagnosis and work details.

Stock: Most parts are bought per job from local suppliers. Oil, filters and bulbs are stocked; counts are reduced manually when used. Unused parts are returned for credit (Q6). Receipt-entry procedures were not described.

Reports: The front desk manager selected repair-history searches by phone number and an open-jobs dashboard showing waiting for parts and ready for pickup (Q8). Report frequency was not specified.

When something changes

Situation	Shop response and requirement for review
Work not approved or cancelled	Work proceeds after approval. Declined-work/cancellation handling and stock adjustments were not described; no further rule is confirmed.
A part is unavailable	No separate missing-part procedure was given. The requested dashboard must show waiting for parts; the visit remains visible as an open job (Q8).
Partial payment or billing correction	No deposits or partial payments. Full cash/card payment at pickup. Closed, paid invoices cannot change; a separate credit/refund records a correction (Q7).
Wrong part delivered	Mechanic reports the error. The front desk manager requests a vendor exchange, notes the delay on the RO and, if needed, moves the car off the lift and calls with a revised pickup time. Record the delay and waiting status for review (Q9).

4 Data and application requirements

Interview answers are confirmed below. Database & App Managers: review the existing design columns against these answers and make any required changes.

Interview topic and source Mae - Done	Data needed Marquise - To do	Application behavior Marcel - To do
Customer / vehicle fields (Q4) Name, phone, optional email; VIN, plate, make/model/year, exact arrival mileage.	customer, vehicle Name/contact; VIN, year, make, model	Add/find customer and vehicle; avoid repeated VINs.
Job information / workflow (Q3, Q4, Q7, Q8) Complaint, diagnosis/notes, approval and mechanic; approver name/time on RO. Show waiting for parts and ready for pickup.	repair_order Complaint, diagnosis, odometer, authorization, status	Open a visit, authorize work, show open jobs, close or cancel.
Labor and parts (Q6) Guide hours x hourly rate; diagnostics use flat hours/hourly pricing. Most parts bought per job.	employee, service, labor_line, part, part_line	Record mechanic, service, hours, quantity and prices charged.
Stock handling (Q6, Q9) Oil, filters, bulbs stocked; counts reduced manually on use. Unused parts returned for credit; wrong parts exchanged.	inventory_movement Part, quantity change, reason, optional job	Receive/use/return stock; prevent negative stock.
Billing rules (Q7) Parts, labor, shop supplies, state tax. Full cash/card payment at pickup; no deposits/partial payments. Paid, closed invoice unchanged; separate credit/refund.	invoice, payment Bill amounts, payment amount, date and method	Issue bill; record allowed payments; calculate remaining balance.
Relationships (Q5) Customer can have several family vehicles; one current owner per vehicle. Relink on sale, retaining history. Usually one mechanic, rarely several. Invoice count not specified.	Vehicle ownership; mechanic per line; invoice per visit	[Confirm or revise the relationship counts in the ERD.]

Three proposed queries

Question Marcel - To do	Input and expected result
What happened to this vehicle?	Filter by VIN; list repair dates, complaints and status, with billed totals when available.
Which jobs are still open?	No input; show open/in-progress orders with vehicle/customer and opening date.
Which bills are still unpaid?	No input; show invoices with total, amount paid and positive remaining balance.

Interviewee priorities (Q8): Change the first query to search customer repair history by phone number, across linked vehicles. Keep open jobs and include waiting for parts and ready for pickup. Unpaid bills were not selected as a priority; full payment is required at pickup.

Other required fields/functions: Review optional email, plate, exact arrival mileage, approval name/time, shop supplies and tax, vehicle history on owner change, and separate credits/refunds. These are requirements for the Database and App Managers to assess, not completed design changes.

Unanswered questions: Brooke stated none (Q10). Cancellation handling, catalog/staff maintenance and invoice counts per visit were not covered; Mae can follow up if a design decision requires an answer. Check the answers against Appendix A.

5 ERD schema and screen examples

Conceptual design and schema

Marquise | To do

Database Manager:

check Database_Guide.pdf against the interview and attach it as Appendix A. Keep both Chen sheets together. Editable diagrams and schema are in the diagram_sources folder beside this report.

Our proposed design contains 11 entities. A customer can have several vehicles, and each vehicle can have several repair visits. Each visit records its bill-to customer separately. Labor and parts are stored as individual lines so a visit can include several services or parts. Each line records what was charged at the time. An invoice belongs to one repair visit, and several payments may apply to that invoice.

Changes made after the interview: [list each changed field, relationship or rule and its reason].

Unchanged assumptions checked with the interviewee: [list].

Keys constraints and normalization

Primary keys identify individual records. Foreign keys connect records without copying the full customer, vehicle or part into each row. A unique VIN avoids duplicate vehicle records, and a unique order link allows at most one invoice per visit in the proposed model. Stock and payment checks prevent negative stock and overpayment. Historical job prices and invoice labels are preserved so later catalog changes do not rewrite old bills.

[Confirm these rules match the interview. Explain any rule removed or changed. State the selected database engine and how its implementation will enforce the rules.]

Application screens

Marcel | To do

Application Manager: add two or three screenshots or screen sketches here.

Screen	Input and output to show
Customer and vehicle	Input: name/contact and VIN/car details. Output: saved linked records or an error. [Insert screenshot/sketch]
Repair and billing	Input: complaint, authorization, labor and parts. Output: repair lines, invoice total and payment balance. [Insert screenshot/sketch]
Report	Input: report choice / vehicle filter. Output: readable results for one of the three queries. [Insert screenshot/sketch]

Required tools: [fill]. Sample currently uses Python and SQLite.

Changes needed before implementation: [fill].

6 Project schedule and meeting log

Report Manager:

Fill real calendar dates and names. The shaded cells below suggest the order of work; move them to fit the actual deadlines.

Report 1 due: [date] Final due: [date]

A [dates]: interview B [dates]: Report 1 C [dates]: build D [dates]: final

Task	Owner	A	B	C	D
Interview and report content	Mae	X	X		
ERD schema and report assembly	DB + Report		X		
Database and app integration	DB + App			X	
Full checks and fixes	Everyone			X	X
Manuals slides and rehearsal	Report + all				X

Concrete handoffs

Interview Manager posts the completed form and report sections by [date].

Database Manager posts the checked diagram/schema by [date].

Application Manager posts screens and setup notes by [date].

Report Manager posts the assembled draft for everyone to review by [date].

Actual meeting and contribution log

Record real meetings or group-chat work: date, place/channel, each person's contribution and next task.

Date time and place or channel	Each person's actual contribution	Decision and next step
[Fill after it happens]	Mae: [fill] Marquise: [fill] Marcel: [fill] Dashaun: [fill]	[What was agreed; task, owner and due date]
[Next actual session]	[Names and contributions]	[Decision and next task]

Before submitting

- ☒ Interview answers and genuine signature attached
- ☐ All placeholders resolved and introduction updated
- ☐ Processes, requirements, ERD and schema agree
- ☐ Screens, dated schedule and accurate log included
- ☐ Everyone checked their section; contents/page numbers updated

Mae | Done

Final checks | To do

Leo's Auto: quick interview

Opening:

"We're doing a small Towson University database class project. Could you explain how you track a repair from arrival to payment? We only need general information, not customer records. May I take notes?"

Date / time

09/29/2026 / 4:45 PM

Place or call/video

In-person at the shop

Student(s) present

Mae Beeman

Person interviewed

Brooke Maenner

Their role

Daughter / Front Desk Manager

1. Quick business profile

Proposed site: Leo's Auto Inc., 14 Willow Ave, Towson, MD 21286. Confirm or correct the name/address below. Approximate answers are fine; sales can be "not disclosed."

History, location(s), staff/size, service area, main services, optional sales

Confirmed: Leo's Auto Inc., 14 Willow Ave, Towson, MD 21286.
Family-owned and operated; founded by Leo. One location, about 5-6 staff.
Family runs the front desk; mechanics handle repairs. Serves Towson and the surrounding area. General auto repair, maintenance, A/C, brakes and tire repair.
No car sales; service only.

2. What tools do you use now? Anything hard to track?

Software / paper / other; one issue or improvement (or "works fine")

Digital shop management software for estimates, plus paper Repair Orders (ROs) for mechanics. Difficulties: finding older paper records for warranty checks and tracking physical part returns to NAPA when the wrong item is delivered.

3. Walk me through a normal repair, from arrival to pickup.

Who does each step? What is written down or entered? Include estimate/approval and payment.

Actual steps, in order

1. Brooke checks in the customer, records details/complaint and prints an RO.
2. Mechanic takes car/RO to bay, diagnoses and writes parts/labor on the RO.
3. Brooke prices parts/labor and calls the customer for estimate approval.
4. After approval, Brooke orders parts; the mechanic completes the repair.
5. Brooke finalizes the invoice in the system.
6. Customer pays in full by cash/card at pickup and receives the keys.

What information do we need?

4. What details do you record?

Customer details (e.g., name, phone, business/fleet account)

Name, phone number and sometimes email. Mainly personal/family vehicles; few fleet accounts.

Vehicle details (e.g., VIN, plate, make/model/year, mileage)

VIN, license plate, make, model, year and exact mileage at arrival.

Repair/job details (e.g., complaint, diagnosis, approval, status, mechanic)

Initial complaint, mechanic diagnosis/notes, customer approval status and assigned mechanic.

5. How do these records connect?

Can one customer have several vehicles? Can a vehicle have several owners/contacts? Can several mechanics work on one job? What happens when a vehicle changes owners?

Answers / exceptions (yes, no, unsure + a short explanation)

One customer may have several vehicles under a family profile. One vehicle is linked to one owner; on sale, create the new customer profile and relink the vehicle, preserving its history. Usually one mechanic per job; rarely several.

6. How do you record labor and parts?

Labor: hours, flat price, or both? Parts: stocked or bought per job? When does stock go down, and what happens to unused/returned parts?

Labor pricing and parts/stock rules

Labor: flat-rate guide hours x hourly rate; diagnostics use flat hours/hourly pricing. Parts usually bought per job from local suppliers. Oil, filters and bulbs stocked; counts reduced manually when used. Unused parts returned for credit.

7. How do approval, bills and payments work?

How is approval recorded? What makes up the total (labor, parts, tax/fees)? Any partial payments, deposits or refunds? Can an old invoice change?

Short answers; note anything we should ask about later

Approval: verbal by phone; name/time written on RO. Total: parts, labor, shop supplies and state tax. No deposits or partial payments; cash/card in full at pickup. Closed, paid invoices cannot change; issue a separate credit/refund.

Finish with priorities and proof

8. What should a small class prototype do first?

Which 2-3 tasks, searches or reports are useful? For example, finding repair history, viewing open jobs or checking unpaid bills. Ask them to choose.

Top tasks / reports

1. Search customer repair history by phone number.
2. Open Jobs dashboard showing status, including waiting for parts and ready for pickup.

9. What happens when something goes wrong?

Ask about a cancellation, declined work, wrong entry or missing part. One common example is enough.

Example and what the shop does

Wrong part delivered: mechanic reports it; Brooke requests a vendor exchange and notes the delay on the RO. If needed, move the car off the lift and call the customer with the revised pickup time.

10. Anything to correct, or a question we should follow up on?

No corrections or additional follow-up questions stated.
Brooke agreed to note-taking for the Towson University class project.

Business name / blank form / photo use permitted? Any restrictions?

Business name use allowed. Exterior/front desk photos discouraged; no plates or PII.

Interview confirmation

This signature confirms that the interview above took place. It does not approve our report, commit the business to using our software, or endorse the project.

Interviewee signature (sign by hand):



Signed electronically

Date signed: 09/29/2026

Matching schema

PK = unique row ID. FK = link to another table. Every table has one integer PK. The ERD relationships supply the FKs below

customer - Contact details for a customer.

PK customer_id; name, phone, email, active

vehicle - A vehicle and its current customer.

PK vehicle_id, FK customer_id, vin, license_plate, model_year, make, model

customer_id links to customer.customer_id.

Also unique: VIN.

repair_order - One visit; bill-to customer is recorded separately from current ownership.

PK order_id, FK vehicle_id, FK bill_to_customer_id, opened_at, complaint, diagnosis,

odometer, status, approval_name, approved_at, authorization_note

bill_to_customer_id links to customer.customer_id; vehicle_id links to vehicle.vehicle_id

employee - Staff; only an active mechanic can be selected for a labor line.

PK employee_id, name, role, active

service - Catalog of labor types and current rates.

PK service_id, name, rate_cents, active

labor_line - A service performed on a job, by one mechanic, at the rate charged.

PK labor_line_id, FK order_id, FK service_id, FK mechanic_id, description,

hours_hundredths, rate_cents

mechanic_id links to employee.employee_id; service_id links to service.service_id;

order_id links to repair_order.order_id

Converting Chen to tables

An entity becomes a table; its ovals become columns. For one-to-many links, put the FK on the many side. For the optional one-to-one order/invoice link, invoice.order_id is a unique FK. LaborLine and PartLine are separate line entities, so each use can store its own quantity, hours, and price.

Schema and simple rules

part - Part catalog; stock is calculated from inventory movements.

PK part_id, part_number, name, price_cents, reorder_level, active

Also unique: part_number.

part_line - Parts used on a job, at the price charged.

PK part_line_id, FK order_id, FK part_id, description, quantity, unit_price_cents

part_id links to part.part_id; order_id links to repair_order.order_id

inventory_movement - Each stock increase or decrease; a repair link is optional.

PK movement_id, FK part_id, FK order_id, quantity_delta, reason, created_at

order_id links to repair_order.order_id (optional); part_id links to part.part_id

invoice - One final bill per order, with labels and amounts preserved.

PK invoice_id, FK order_id, issued_at, customer_label, vehicle_label, labor_cents, parts_cents, shop_supplies_cents, tax_cents

order_id links to repair_order.order_id

Also unique: order_id.

payment - A payment that settles the outstanding balance of an invoice.

PK payment_id, FK invoice_id, paid_at, amount_cents, method, request_key

invoice_id links to invoice.invoice_id

Also unique: request_key.

Rules already in the sample

Money is stored as integer cents; hours as hundredths. Stock cannot go below zero.

Payments must be positive and must equal the outstanding balance. VINs cannot repeat. Only authorized, in-progress orders accept work lines. Final orders and issued bills are locked.

Confirm before keeping these choices

One current customer per vehicle; one mechanic per labor line; at most one final invoice per visit. No deposits before invoicing, refunds, invoice corrections or shared vehicle ownership yet. Tax is a manually entered demo amount. Change the model if the interview requires it.

Why separate tables?

Customer details, service rates and part prices each have one home. Job lines preserve the rate actually charged; invoice labels preserve the billed details. These are historical facts, not copies to keep synchronized. Stock = sum of movements; balance = invoice total minus payments.